



TUTORIAL PROPOSAL

Title

Error-Aware Socratic Tutoring at Scale: Personalized and Safe Human-AI Dialogue in K12 Math

Main Organizer

Tianlong Xu

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- Short Bio: Tianlong Xu, Ph.D., is a Staff Applied Scientist at Squirrel AI. Prior to this, he worked as a machine learning scientist at Goldman Sachs and Home Depot. He has rich experience in building and productizing machine learning models, ranging from RAG powered by LLMs, option pricing & sales, anomaly detection, recommender systems, etc. He has published five papers in top AI conferences like KDD, AAAI and has received IAAI Deployed Application Awards at AAAI25. He is a Google Cloud certified Generative AI Leader. He holds five patents in the U.S. and China, and Ph.D. degree from Georgia Institute of Technology, USA.

Qingsong Wen

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- Short Bio: Qingsong Wen is currently the Head of AI & Chief Scientist at Squirrel Ai Learning (a top EdTech unicorn), and PhD Supervisor at University of Oxford. He received his M.S. and Ph.D. degrees in Electrical and Computer Engineering from Georgia Institute of Technology, USA. His research interests include machine learning, data mining, and signal processing, especially AI for Time Series, AI for Education, LLM & AI Agent. He has published over 170 top-ranked AI conference and journal papers. Currently, he serves as Associate Vice President for Cybernetics of IEEE SMC Society, Chair of IEEE CIS Task Force on AI for Time Series and Spatio-Temporal Data, and Vice Chair of INNS AI for Education Section. He also regularly serves as Area Chair of the top conferences including NeurIPS, ICML, ICLR, KDD, IJCAI, ICASSP, etc, and Associate Editor for IEEE Transactions on Pattern Analysis and Machine Intelligence (IF=18.6).

All-organizers

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Duration

Half Day

Structure

This half-day tutorial is divided into five sections, each addressing a key component of Socratic AI Tutoring for K12 mathematics. The session blends lectures, hands-on exercises, case studies, and discussion, offering attendees both conceptual depth and practical experience in building responsible, personalized AI-powered education systems.

1. Error-Aware Information Retrieval: AI-Powered Analysis of Student Errors

(45 minutes: 25 min lecture, 20 min hands-on)

- How AI analyses student drafts to detect *where and why* students struggle during math problem-solving
- Common math error types and how AI can pre-cache error signatures to tailor feedback
- Challenges in early-stage detection: hallucinations, multimodal limitations, and how multi-pass refinement improves accuracy



- Hands-on: Use curated student data to experiment with automated error attribution and visualization tools

2. Designing Socratic-Style AI Prompts for Interactive Conversations

(40 minutes: 20 min lecture, 20 min hands-on)

- How to construct AI prompts that encourage step-by-step reasoning and student-led thinking
- Techniques for chaining thought processes, avoiding direct answer delivery, and encouraging exploration
- Hands-on: Practice writing Socratic prompts and test them with sandboxed student problem examples

3. Early-Stopping Mechanisms and Conversational Evaluation

(45 minutes: 25 min lecture, 20 min hands-on)

- Rules for terminating unproductive or harmful AI-student dialogues using language, behaviour, and engagement signals
- Detailed overview of our 6-dimension evaluation framework to rate AI tutoring conversations: correctness, encouragement, precision, efficiency, completeness, and student intent
- Rules for aligning LLM outputs with the developmental and cognitive needs of students across different grade levels
- Techniques to detect and respond to student disengagement, toxic input, and AI hallucinations in real time
- Hands-on: Evaluate real conversation transcripts, practice assigning evaluation scores, and simulate early-stopping triggers using provided moderation criteria

4. Case Studies and Real-World Deployment Lessons

(30 minutes: 10 min presentation, 20 min discussion)

- Showcase of high-impact cases where Socratic AI successfully addressed student misconceptions
- Highlight failure cases: misalignment, misattribution, or over-guidance
- Discussion: What worked, what failed, and how design choices impacted learning outcomes

5. Toward Scalable, Ethical, and Multimodal AI Tutoring

(20 minutes: Lightning talk + group brainstorming)



- Emerging directions: image-based draft parsing, real-time classroom integration, and LLM tuning strategies
- Ethical considerations: bias, over-reliance, and consent in student-AI interaction
- Group brainstorm: How can we scale error-aware Socratic AI across subjects, languages, and learning levels?

Outline

As AI becomes increasingly integrated into K12 education, the challenge is not only to make it intelligent—but also adaptive, instructive, and responsible. This tutorial presents Socratic AI Tutoring, an interactive framework for guiding students through math problem-solving via step-by-step, error-aware dialogue. Rather than simply giving answers, our AI system first analyzes student drafts to determine where and why they got stuck, enabling conversations that are personalized, targeted, and grade-appropriate.

A key innovation lies in combining error attribution, Socratic prompting, and conversational safeguards into one unified system. We also introduce grade-level alignment strategies, a six-dimensional evaluation framework, and moderation mechanisms to promote safe and meaningful dialogue while avoiding hallucinations or student disengagement.

This tutorial aligns with IEEE SMC 2026 themes in Human-Machine Systems and Cybernetics, offering a real-world case study in how AI can enable adaptive, ethical human-machine collaboration in education. Through lectures, hands-on sessions, and case discussions, participants will gain actionable insights into building next-generation AI tutoring systems that are both scalable and socially responsible.

Motivation

As AI continues to shape the future of K12 education, there's a growing need for systems that engage students in meaningful, adaptive learning—rather than acting as static answer generators. Most AI tutoring systems today lack the ability to truly understand why students struggle, resulting in generic and often unhelpful guidance.

Our approach introduces two key innovations:

1. First-of-its-Kind Error-Aware Conversations – By analyzing students' solution drafts prior to dialogue, our system can identify where and why a student made a mistake. This enables AI to deliver targeted, in-depth, and actionable support tailored to each student's learning needs.



2. Socratic, Step-by-Step Reasoning – The AI interacts through structured, back-and-forth guidance that encourages students to think critically, promoting deeper conceptual understanding rather than passive answer consumption.

Aligned with Human-Machine Systems and Cybernetics, this tutorial offers insights into how AI can support safe, intelligent, and grade-appropriate tutoring at scale. We propose this tutorial to share practical methods in error detection, prompt design, conversation moderation, and AI evaluation, helping attendees design more responsible and effective AI-powered educational systems.